

The resilience of business ecosystems in the post-COVID-19 era: Notes on the European Recovery and Resilience Facility

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Abstract

Today's COVID-19 crisis drives various socioeconomic systems to a profound readjustment and restructuring. This presentation aims to examine the need to reinforce the resilience of business ecosystems, suggesting future research avenues mainly at the regional level. In terms of design, a synthesis of the resilience and business ecosystems' concepts is approached, followed by introducing essential elements concerning the European Recovery and Resilience Facility (RRF). The conclusions are twofold. First, the resilience of business ecosystems is a significant development pillar for the post-COVID-19 era, even though the RRF does not include provisions for regional development (the national authorities are responsible). Second, the resilience of local and regional authorities is significant, especially in areas where small and medium-sized enterprises (SMEs) rely on activities affected by the pandemic, such as tourism and transport. Further research could focus on the relationship between SME innovation and critical infrastructure.

Keywords: COVID-19 crisis, business ecosystems, resilience, Recovery and Resilience Facility

1. Introduction

Since the first days of 2020, the COVID-19 crisis (hereafter pandemic) has led the global socioeconomic system to a profound transformation and restructuring. According to various analysts, today's crisis of globalization caused by the pandemic drives the acceleration of the fourth industrial revolution and the increasing "blurring" between the digital and physical world (Bonilla-Molina, 2020; Czifra & Molnár, 2020).

The social distancing and lockdown measures have contributed to a multilevel readjustment of the labor and business environment—for both the developed and less resilient socioeconomic systems (ILO, 2020; UNCTAD, 2020; World Economic Forum, 2020). According to the Food and Agriculture Organization (FAO, 2020), rural areas will face the greatest danger since (in

most cases) the population was already living in extreme poverty conditions—a trend exacerbated by the pandemic.

In this context, it seems that all European regions must adapt to the new environment by retaining and reinforcing their competitive advantages (Henry & Smith, 2021; Mason & Hruskova, 2021). More specifically, according to Epifanova et al. (2020), the post-COVID-19 era necessitates the generation of innovative regional enterprise ecosystems. A significant move towards this direction is the establishment of the Next Generation EU (NGEU), a subsidy program of enormous proportions for the European Union (EU) member states, focusing on the goals of digital transformation, green energy (“European Green Deal”), and resilience. According to the conclusions from the Special meeting of the European Council held between 17 and 21 July 2020 (European Council, 2020), these programs must prioritize strengthening the most affected sectors and industries since the crisis is structural and multilayered.

This presentation acknowledges this need for enhancing the resilience of regional business ecosystems and suggests future research avenues in the causes of relative underdevelopment. This introductory and conceptual study of the resilience of business ecosystems aims to end up with a reframed theoretical context by elliptically exploring the related literature of resilience (Jaakkola, 2020; Snyder, 2019). Specifically, in section two, various elements of the “business ecosystems resilience” are investigated. In the third section, considering these recent theoretical developments, the EU Recovery and Resilience Facility’s (RRF) main elements are presented. In the fourth and last section, a synopsis is made, suggesting future research directions.

2. What makes a business ecosystem resilient?

The answer to this question requires first the clarification of the concept of the business ecosystem. The “entrepreneurial ecosystem” or “business ecosystem” are two concepts used by the recent scholarly literature interchangeably. However, there are differences. The business ecosystem refers to the relationships between the business environment and the institutional and socioeconomic context, while the entrepreneurial ecosystem concerns the supporting structures, such as incubators, startup accelerators, and entrepreneurial education (Burda et al., 2020).

By exploring the contributions of Marshall (1890)¹ and Veblen (1898), various economists attempted to interpret the development of socioeconomic systems through the prism of biology. Via significant developments over the twentieth century in the fields of evolutionary economics and strategic management (Chatzinikolaou & Vlados, 2019), the related literature treats as a

¹ A. Marshall, the founder of Neoclassical Economics, argued that the “The Mecca of the economist lies in economic biology” (Hodgson, 1993).

milestone Moore’s work (Moore, 1993), who introduced the business ecosystem concept with the seminal article “Predators and Prey.” More specifically, Moore underlined the contiguity between the biological and business ecosystems, arguing that today’s business world is the outcome of “organisms” that generate co-evolving communities with shared values. In a similar vein and with equal treatment by the related scholarly literature, Iansity and Levien (2004) postulated that business ecosystems are primarily the outcome of their stakeholders’ strategy, with the “keystone species” playing the most decisive role for the ecosystem’s survival and sustainability. From a more generic perspective, various analysts classified next the components that make up a business ecosystem. Isenberg (2016) offered one of the most comprehensive relevant approaches, suggesting the six domains that lead the business ecosystem to self-sustainment—finance, culture, human capital, markets, policy, and supports.

Today, the contributions in this literature are increasing, which derive from diverse scientific fields (Alvedalen & Boschma, 2017). In this context, the resilience of business ecosystems has risen significantly over the past years. Holling’s work (Holling, 1973, p. 17) on the stability and resilience of ecological system constitutes one of the fundamentals of the field, arguing that resilience means the “*ability of these systems to absorb changes of state variables, driving variables, and parameters, and still persist.*” Next, the growing study of the concept led to examining adjacent perspectives (most notably ecology) that contributed to resilience’s popularity, nowadays considered one of the necessary features for localities, individuals, and firms (Meerow & Newell, 2015). According to Zhu and Ruth (2013, p. 74), even the “industrial ecosystems” have the attribute of resilience, in the sense of their ability to sustain “*their defining feature of eco-efficient material and energy flows under disruptions.*” Table 1 presents significant recent definitions of the resilience of business ecosystems in the related literature.

Table 1: Significant recent definitions in the “business ecosystems resilience” literature

Author	Definition of the resilience of the business or entrepreneurial ecosystem (EE)
Roundy et al. (2017, p. 99)	“We define [resilience] as the degree to which an EE can continuously recover from and adapt to exogenous shocks and endogenous pressures.”
Ramezani and Camarinha-Matos (2019, p. 44)	“In fact, the innovations associated with the 4th industrial revolution need to be embedded disruptions-coping strategies, namely: (1) to enhance the ability of the business ecosystems to prepare for and adapt to disruptive market environments and withstand and recover rapidly from the impacts of unforeseen disasters, and (2) to stay competitive in volatile business environments. Therefore, the characteristics of resilience and antifragility including responsiveness, learning, adaptability, etc. need to be included in the design of advanced cyber-physical systems.”

Sohns and Wójcik (2020, p. 1542)	<i>“Resilient EEs are characterized by high entrepreneurial diversity. Therefore, their ability to attract external actors, while preventing their loss through outmigration or failure, is crucial for the resilience of the EE. In addition, resilient EEs are also characterized by actors’ ability to create and maintain internal and external ties, which increases actor connectivity and, consequently, fosters knowledge exchange and the recycling of resources.”</i>
Iacobucci and Perugini (2021, p. 6)	<i>“It is likely that the recovery phase implies some kind of adaptation and internal modification of local systems; indeed, the strength of the EE, which we consider as the main explanatory variable of resilience, should imply the ability of entrepreneurs and institutions not only to resist but also to react and innovate.”</i>

These various definitions converge to the dynamic created within the local system. At the same time, the approach to innovation is remarkable, conceptualized as the necessary adaptation mechanism for the newly emerging fourth industrial revolution conditions. Also, the literature concerning the resilience for navigating through the current pandemic crisis proliferates with an increasing pace—primarily referring to micro-level implications. Various relevant contributions examine the role of entrepreneurial ecosystems to foster startups and economic recovery through regional and institutional innovation, based on training programs for resilient entrepreneurs (Maritz et al., 2020). From this perspective, universities and public policymakers supporting the entrepreneurial ecosystem constitute the most significant actors in this innovation effort (Portuguez Castro & Gómez Zermeño, 2020). Also, the literature particularly emphasizes the need for entrepreneurial mindset and creativity on the part of small and medium-sized enterprises (SMEs) to exit the crisis, which depends on the entrepreneur’s skills and the ecosystem that facilitates this effort with the infrastructure provided (Ratten, 2020; Thukral, 2021).

Overall, it seems that resilience is today a prominent field of study, concerned with the factors supporting the long-term development of a business ecosystem. At the same time, the business ecosystem concept is a dynamic notion, compared to the static and traditional concept of regional development, helping to clarify the domains for potential integrated and structural development intervention instead of focusing on mere growth. The following section examines elements of the actions undertaken by the EU to reinforce its resilience in the post-COVID-19 era.

3. The European Recovery and Resilience Facility

As mentioned, the EU leaders agreed in mid-2020 to promote resilience through the NGEU. This emergency package and “common debt” among the member states will invest 750 billion euros in the green, digital, and resilient Europe—390 billion in grants and 360 billion in loans.

The RRF is the core of the NGEU since it involves 75% of the funds in the Commission's original proposal—increased to 672.5 billion by the European Council (Crum, 2020).

Therefore, the EU promotes resilience significantly, suggesting this attribute as the primary ability to exit today's crisis. However, the RRF does not have a clear territorial focus and does not mention the need to involve the local and regional authorities in the national recovery plans for the post-COVID-19 era (Conference of Peripheral Maritime Regions, 2020). The RRF, though, emphasizes SMEs, which are more vulnerable and less resilient than larger firms. The RRF acknowledges that SMEs have a limited ability to diversify their activities if their sector shuts down temporarily. Also, the primary problem for European SMEs is their involvement in sectors affected most by the lockdown measures, such as tourism and transport (OECD, 2020).

Furthermore, the pandemic crisis surfaced the shortcomings of specific European regions in digital infrastructure, making clear a divide between the development perspectives of different regions and cities. In this context, as suggested by the 2020 EU regional and local barometer (European Committee of the Regions, 2020), local and regional authorities must address the regulatory barriers and offer financial incentives for digital transformation. Also, as this report mentions, the structure and vulnerability of the various European regions must constitute the essential criteria for distributing the recovery and resilience funds, focusing on the risk associated with their reliance on tourism, international trade and transport, the number of self-employed and small firms, and the GDP loss since the crisis started. These regions must broaden their local economies and not rely on a single sector. This report concludes that public authorities must offer significant incentives for resilience, which derive from digitalization, the principles of cyclical economy, and sustainability (Chatzinikolaou et al., 2021).

Therefore, the case of Europe reaffirms the principles suggested by the literature on the resilience of business ecosystems. The RRF does not emphasize specific spatial structures since it focuses on the national recovery plans, whose implementation is the primary way to improve regions' resilience.

4. Final remarks and future research avenues

This research presented the fundamental features of the concept of business ecosystems, concluding that resilience is a significant pillar of development in the post-COVID-19 era. The presentation of essential elements of the Recovery and Resilience Facility showed no extended vision on the territories affected by the pandemic (local and regional authorities), making national governments responsible for including business ecosystems in their policies. However, the design that strengthens the resilience of local and regional authorities is imperative since various regions face increased risks due to their reliance on activities affected by the

acceleration of the fourth industrial revolution and the pandemic (tourism, international trade, transport, the self-employed, and SMEs in generic terms).

This introduction to the resilience of business ecosystems could specialize in studying specific sectors, aiming to find how the various business ecosystems (regions) adapt to the new conditions and how resilient they are. It seems that the resilience of a business ecosystem depends on the development of critical infrastructure, including transport, tourism, and health (Dimitriou & Sartzetaki, 2020). In a similar vein, future research could distinguish how the business ecosystem evolves by focusing on the relationship between the innovation of SMEs and the existence of critical infrastructure. Since we are based on the less developed (and less resilient) Eastern Macedonia and Thrace region, action research to learn more about this correlation in the various regional actors could be possible.

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